

Assignment 1: Student Course Registration Portal

Problem Statement

Design and develop a simple **Student Course Registration Portal** using Java Servlets, HTML, and JSP. The application should allow a student to register for a course by submitting personal and course-related details. The system should validate the submitted details and either show a confirmation page or return the student to the registration form.

This assignment is intended to test Servlet fundamentals, form handling, Servlet lifecycle methods, server-side forwarding, and client-side redirection.

Functional Requirements

The application should contain a home page that welcomes the user to the Student Course Registration Portal. From the home page, the user should be able to open the course registration form.

The registration form should collect the following details:

Field	Description
Student Name	Name of the student
Email	Student email address
Age	Student age
Course Name	Course selected by the student
Preferred Batch Time	Student's preferred batch timing

The available course options should include at least four courses, such as:

Course Options

Java Full Stack
Python Full Stack
MERN Stack
Data Analytics

After submission, the system should validate the entered data.

Validation Rules

The registration should be considered valid only if:

Field	Validation Rule
Student Name	Should not be empty
Email	Should not be empty
Age	Should be 18 or above
Course Name	Should be selected
Preferred Batch Time	Should be selected or entered

Expected Application Flow

If the entered details are valid, the user should be taken to a confirmation page displaying all submitted details.

If the entered details are invalid, the user should be redirected back to the registration form.

The confirmation page should display:

Details to Display
Registration Successful message
Student Name
Email
Age
Course Name
Preferred Batch Time

Servlet Requirements

The application must include a Servlet responsible for processing the registration request.

The Servlet should demonstrate the use of:

Concept	Expected Usage
Servlet lifecycle	Initialization, request processing, and destruction
Request handling	Reading form data submitted by the user
RequestDispatcher	Forwarding valid data to the confirmation page
sendRedirect	Redirecting invalid submissions back to the form
Request attributes	Passing valid data from Servlet to JSP

Assignment 2: Employee Leave Application System

Problem Statement

Design and develop an **Employee Leave Application System** using Java Servlets, HTML, and JSP. The system should allow an employee to apply for leave by entering leave-related details. Based on the entered data, the system should validate the request and show a leave review page if the request is valid.

This assignment is intended to evaluate the student's understanding of Servlet request processing, lifecycle methods, validation logic, RequestDispatcher, and sendRedirect.

Functional Requirements

The application should provide a leave application form for employees.

The leave form should collect the following details:

Field	Description
Employee Name	Name of the employee
Employee ID	Unique employee identifier
Department	Employee department
Leave Type	Type of leave requested
Number of Leave Days	Number of days requested
Reason	Reason for leave

The leave type should include options such as:

Leave Type Options

Sick Leave
Casual Leave
Emergency Leave
Work From Home

After submitting the form, the system should process and validate the leave application.

Validation Rules

The leave request should be considered valid only if:

Field	Validation Rule
Employee Name	Should not be empty
Employee ID	Should not be empty
Department	Should not be empty
Leave Type	Should be selected
Number of Leave Days	Should be between 1 and 10
Reason	Should contain at least 10 characters

Business Rules

The system should apply the following business logic:

Condition	Message to Display
Leave days are greater than 5	This leave request requires manager approval
Leave days are 5 or fewer	This leave request can be processed normally

Expected Application Flow

If the leave request is valid, the user should be forwarded to a leave review page.

If the leave request is invalid, the user should be redirected back to the leave form.

The leave review page should display:

Details to Display
Employee Name
Employee ID
Department
Leave Type
Number of Leave Days
Reason
Approval message

Servlet Requirements

The application must include a Servlet responsible for processing the leave application.

The Servlet should demonstrate the use of:

Concept	Expected Usage
Servlet lifecycle	Show clear initialization, processing, and destruction behavior
Request handling	Read all leave form data
Validation	Validate employee and leave details
Business logic	Decide whether manager approval is required
RequestDispatcher	Forward valid leave applications to the review page
sendRedirect	Redirect invalid leave applications back to the form
Request attributes	Pass processed leave data to the JSP page

Learning Outcomes

After completing this assignment, students should be able to:

Learning Outcome

- Build a form-based Servlet application
- Process employee form data using Servlets
- Apply validation and basic business rules
- Use RequestDispatcher for server-side navigation
- Use sendRedirect for client-side navigation
- Pass data from Servlet to JSP
- Understand the role of lifecycle methods in Servlet execution

Assignment 3: Online Exam Result Preview System

Problem Statement

Design and develop an **Online Exam Result Preview System** using Java Servlets, HTML, and JSP. The system should allow a student to enter basic details and marks for three subjects. The application should calculate the total marks, percentage, and result status. If the entered data is valid, the result should be displayed on a preview page. If the data is invalid, the user should be redirected back to the exam form.

This assignment is intended to test Servlet fundamentals, request processing, validation, arithmetic logic, lifecycle methods, RequestDispatcher, and sendRedirect.

Functional Requirements

The application should contain an exam result form.

The form should collect the following details:

Field	Description
Student Name	Name of the student
Roll Number	Student roll number
Subject 1 Marks	Marks obtained in first subject
Subject 2 Marks	Marks obtained in second subject
Subject 3 Marks	Marks obtained in third subject

Each subject should be evaluated out of 100 marks.

Validation Rules

The result should be calculated only if:

Field	Validation Rule
Student Name	Should not be empty
Roll Number	Should not be empty
Subject Marks	Should be numeric
Subject Marks	Should be between 0 and 100

If any validation fails, the user should not be shown the result preview page.

Result Calculation Rules

The system should calculate:

Calculation

Total marks

Percentage

Result status

The result status should be decided using the following rules:

Percentage

75% and above

60% to below 75%

40% to below 60%

Below 40%

Result Status

Distinction

First Class

Pass

Fail

Expected Application Flow

If all entered details are valid, the user should be forwarded to a result preview page.

If any entered detail is invalid, the user should be redirected back to the exam form.

The result preview page should display:

Details to Display

Student Name

Roll Number

Subject 1 Marks

Subject 2 Marks

Subject 3 Marks

Total Marks

Percentage

Result Status

Servlet Requirements

The application must include a Servlet responsible for processing the result calculation request.

The Servlet should demonstrate the use of:

Concept

Servlet lifecycle

Request handling

Validation

Business logic

RequestDispatcher

sendRedirect

Request attributes

Expected Usage

Initialization, request handling, and destruction

Reading student details and marks

Checking empty, invalid, or out-of-range input

Calculating total, percentage, and result status

Forwarding valid result data to JSP

Redirecting invalid input back to the form

Passing calculated result data to the JSP page